

TOP NEWS TIMES

2026년 7월 27일 ~ 8월 2일 · 제31주호

On Air · Daniel

10편 수록



이번 호 기사

01	환경	Cities Around the World Are Running Out of Groundwater	2
02	경제	Global Race to Find Chip Engineers Heats Up	3
03	환경	Nuclear Power Makes a Quiet Comeback	4
04	세계	Nations Restrict Grain Exports, Raising Global Food Concerns	5
05	기술	The High-Stakes Debate Over Regulating AI	6
06	기술	The Race to Fill the Sky With Satellites	7
07	사회	Robots Step In to Care for the Elderly	8
08	경제	Europe's Carbon Tax Reshapes Global Trade	9
09	건강	The Silent Pandemic Hiding in Plain Sight	10
10	환경	The Hidden Water in Your Food	11

Cities Around the World Are Running Out of Groundwater

By Daniel · 2026-07-30 · 133 words · L1



도시 지하수 고갈

Groundwater — water stored deep beneath the soil — supplies nearly half the world's drinking water. But in many cities, this resource is disappearing faster than it can be replaced. Rapid urban growth, heavy farming, and years of drought are all making the problem worse.

In cities like Jakarta, Mexico City, and Chennai, the ground is actually sinking because so much water has been pumped out from below. Scientists call this process 'land subsidence,' and

it can damage roads, buildings, and drainage systems. Some areas have dropped by several meters over recent decades.

Several cities are already taking action. Bangkok and Tucson have introduced programs to collect rainwater and return it to underground reserves. Experts say that reducing waste and managing water more carefully are the most important steps cities can take now.

한글 요약

전 세계 식수의 절반가량을 공급하는 지하수가 도시 성장, 농업, 가뭄 등으로 인해 빠르게 고갈되고 있다. 자카르타, 멕시코시티, 첸나이 등 여러 도시에서는 과도한 지하수 취수로 인한 지반 침하 현상이 심각하게 나타나고 있으며, 도로와 건물 등 기반 시설에 피해를 주고 있다. 방콕과 투손은 빗물을 지하 수층으로 환원하는 프로그램을 도입하는 등 일부 도시들이 적극적으로 대응에 나서고 있다. 전문가들은 물 낭비 절감과 철저한 수자원 관리가 현재 도시들이 취할 수 있는 가장 핵심적인 조치라고 강조한다.

Source · theconversation.com

WORDS & MEANINGS

groundwater 지하수

resource 자원

drought 가뭄

subsidence 침하, 땅이 가라앉는 현상

pump 펌프로 퍼올리다

reserves 비축량, 저장량

urban 도시의

Global Race to Find Chip Engineers Heats Up

By Daniel · 2026-07-30 · 143 words · L1



세계 반도체 인재 쟁탈전

Countries and companies around the world are competing fiercely to attract semiconductor engineers. Chips power everything from smartphones to electric cars, and demand for them keeps growing. As a result, skilled workers in this field are in very short supply.

The United States, South Korea, Japan, and the European Union are all investing heavily in domestic chip production. To support these factories, they need thousands of trained

engineers and technicians. Many experts say the industry will need far more workers than are currently available in the coming years.

The U.S. has placed restrictions on sending semiconductor technology and talent to China, which has pushed China to train its own engineers at home. Meanwhile, companies like TSMC are opening facilities in the U.S. and Japan, partly to access local talent pools. The battle for chip workers is reshaping where and how the industry grows.

한글 요약

전 세계적으로 반도체 수요가 급증하면서 미국, 한국, 일본, 유럽연합 등 주요국이 자국 내 반도체 생산에 대규모 투자를 집행하고 있다. 그러나 이를 뒷받침할 숙련 엔지니어와 기술자는 심각하게 부족한 상황이며, 전문가들은 향후 인력 수급 불균형이 더욱 심화될 것으로 전망한다. 미국의 대중국 반도체 기술·인재 유출 제한 조치는 중국이 독자적인 인재 육성 체계를 구축하도록 압박하고 있다. TSMC 등 주요 기업들이 현지 인재 확보를 위해 미국과 일본에 생산 시설을 설립하는 등, 반도체 인력 확보 경쟁은 글로벌 반도체 산업의 지형을 새롭게 재편하고 있다.

WORDS & MEANINGS

semiconductor 반도체

compete 경쟁하다

shortage 부족

domestic 국내의

restriction 제한, 규제

facility 시설

talent 인재, 재능 있는 사람

Nuclear Power Makes a Quiet Comeback

By Daniel · 2026-07-29 · 297 words · L3



에너지 전환의 딜레마 — 원자력의 귀환

For decades, nuclear energy fell out of favor as concerns about safety and waste dominated public debate. Now, governments and energy companies around the world are taking a second look, driven by rising electricity demand and the urgent need to cut carbon emissions.

Nuclear reactors generate electricity by splitting uranium atoms in a process called fission, which releases enormous amounts of heat. That heat produces steam, which spins turbines to generate power — without burning fossil fuels and therefore without direct carbon emissions.

One of nuclear energy's strongest advantages is that it provides steady, reliable power around the clock,

regardless of weather conditions. This is known as baseload power, and it is something that solar panels and wind turbines cannot consistently deliver on their own.

Renewable energy has grown rapidly over the past decade, and its costs have dropped significantly. However, grid operators still struggle to balance supply and demand when the sun sets or the wind drops, making a reliable backup source essential.

Several countries are responding by extending the lives of existing reactors or building new ones. France has announced plans to construct six new reactors, while Japan has restarted plants shut down after the 2011 Fukushima disaster. In the United

States, technology companies seeking carbon-free power for energy-intensive data centers have signed agreements to support nuclear projects.

Critics argue that nuclear plants take too long to build and cost far too much to be a practical near-term solution. They also point to the unsolved problem of storing radioactive waste safely for thousands of years.

Support for smaller reactor designs, known as small modular reactors, is growing as a possible answer to these challenges. Whether they can be deployed fast enough to make a real difference remains a central question in the global energy debate.

한글 요약

탄소 배출 감축과 전력 수요 증가를 배경으로 세계 각국이 수십 년간 외면받아온 원자력 에너지를 재평가하고 있다. 원자력은 화석 연료 없이 24시간 안정적인 기저부하 전력을 공급할 수 있어, 간헐적인 재생에너지의 단점을 보완하는 수단으로 주목받고 있다. 프랑스의 신규 원자로 건설, 일본의 원전 재가동, 미국 기술 기업들의 원전 지원 협약 등 구체적인 움직임이 이어지고 있다. 그러나 높은 건설 비용과 긴 공사 기간, 방사성 폐기물 처리 문제가 여전히 과제로 남아 있으며, 소형 모듈 원자로(SMR)가 이에 대한 해결책이 될 수 있을지 주목된다.

WORDS & MEANINGS

fission 핵분열

emission 배출(물)

baseload 기저 부하 (안정적 기본 전력 공급)

grid 전력망

radioactive 방사성의

modular 모듈식의, 조립식의

reactor 원자로

Nations Restrict Grain Exports, Raising Global Food Concerns

By Daniel · 2026-07-29 · 288 words · L3



글로벌 식량 안보와 곡물 수출 제한

When a major food-producing country limits its exports, the effects are felt far beyond its borders. In recent years, more countries have been restricting grain exports to protect their own food supplies, and global food prices have risen sharply as a result.

In 2023, India — one of the world's largest rice exporters — banned exports of non-basmati white rice. This decision alone pushed global rice prices to their highest levels in over a decade.

Around the same time, Russia ended its participation in a deal that had allowed Ukraine to safely export grain by sea. Ukraine is one of the top wheat and sunflower oil suppliers in the world,

and the breakdown of that agreement disrupted supplies to countries across Africa, the Middle East, and Asia.

Export restrictions are not new, but their use has increased during recent global crises. Countries impose these limits to keep domestic prices stable and protect their own populations from shortages. However, critics argue that wealthy nations can manage these shocks more easily, while poorer countries that depend on imports suffer the most.

The United Nations Food and Agriculture Organization has warned that food export restrictions make global food insecurity worse. When

several countries act at once, prices rise quickly and low-income nations struggle to buy enough food.

The situation also has a political side. Some analysts note that food has become a tool of geopolitical influence, with governments using access to grain and other staples to strengthen their positions in international relations.

Experts say long-term solutions require greater investment in local food production, stronger international cooperation, and trade rules that prevent countries from cutting off supplies during a crisis. Without these, vulnerable nations will continue to face serious food security risks.

한글 요약

인도의 백미 수출 금지와 러시아의 흑해 곡물 협정 탈퇴 등 주요 식량 생산국들의 수출 제한 조치가 잇따르면서 세계 식량 가격이 급등하고 있다. 이러한 조치는 자국민 보호를 명분으로 하지만, 수입 의존도가 높은 저소득 국가들이 가장 큰 피해를 입는 구조적 불평등을 심화시킨다. 유엔 식량농업기구는 복수의 국가가 동시에 수출을 제한할 경우 전 세계 식량 불안이 급격히 악화된다고 경고하며, 일부 전문가들은 식량이 지정학적 영향력의 수단으로 활용되고 있다고 분석한다. 전문가들은 이 문제의 근본적 해결을 위해 지역 식량 생산 투자, 국제 협력 강화, 그리고 위기 시 수출 차단을 방지하는 국제 무역 규범의 수립이 필요하다고 강조한다.

WORDS & MEANINGS

export 수출(하다)

restriction 제한, 규제

shortage 부족, 결핍

staple 주식, 기본 식료품

geopolitical 지정학적인

food security 식량 안보

disrupted 방해된, 차질이 생긴

The High-Stakes Debate Over Regulating AI

By Daniel · 2026-07-29 · 280 words · L3



AI 규제, 혁신과 안전 사이

Artificial intelligence is changing the way people work, communicate, and make decisions. But as AI systems become more powerful, governments around the world are asking a difficult question: how do you keep this technology safe without slowing down progress?

The European Union passed the AI Act in 2024, making it the first major law in the world to regulate AI in a broad and structured way. The law sorts AI systems into risk categories. High-risk uses — such as AI in hospitals, courts, or hiring processes — must meet strict safety requirements before they can be used.

Supporting the law, EU officials argue that people should be able to trust the AI systems that affect their lives. They say clear rules protect citizens from harm and build public confidence in the technology.

However, some tech companies and industry groups warn that heavy regulation could push innovation elsewhere. If compliance costs are too high, smaller companies may struggle to compete, and new AI products could take longer to reach the market.

The United States has taken a different approach, relying mostly on voluntary guidelines rather than strict legislation. China, meanwhile, has

introduced targeted rules for specific AI applications, especially those related to content generation.

Experts point out that no single approach is clearly the best. Loose regulation can allow harmful AI to spread quickly, while overly strict rules can slow research and give less regulated countries a competitive advantage.

The debate reflects a broader challenge: technology often moves faster than the laws designed to govern it. How governments strike that balance will shape not only the future of AI, but also how fairly and safely it is used by people everywhere.

한글 요약

유럽연합은 2024년 AI 시스템을 위험 등급별로 분류하고 고위험 용도에 엄격한 안전 요건을 부과하는 세계 최초의 포괄적 AI 규제법인 'AI 법'을 제정했다. 반면 미국은 자율적 가이드라인 중심의 접근 방식을, 중국은 콘텐츠 생성 등 특정 분야에 대한 표적 규제를 도입하는 등 각국의 규제 방식은 서로 다르다. 느슨한 규제는 유해 AI의 확산을 초래할 수 있고, 지나치게 엄격한 규제는 혁신을 저해하고 국제 경쟁력을 약화시킬 수 있어, 어느 한 방식이 명백히 최선이라 보기 어렵다. 기술이 법률보다 빠르게 발전하는 현실 속에서, 각국 정부가 규제와 혁신 사이의 균형을 어떻게 설정하느냐가 AI의 미래와 그 공정하고 안전한 활용 방식을 결정짓게 될 것이다.

WORDS & MEANINGS

regulate 규제하다, 통제하다

legislation 법률, 입법

compliance 준수, 법규 이행

category 범주, 분류

voluntary 자발적인

competitive 경쟁력 있는

innovation 혁신

The Race to Fill the Sky With Satellites

By Daniel · 2026-07-29 · 272 words · L2



우주 인터넷 경쟁 — 위성이 하늘을 채운다

Hundreds of satellites are now circling Earth at low altitude, and the number is growing fast. Companies like SpaceX, Amazon, and several others are competing to build what are called megaconstellations — networks of thousands of satellites working together to deliver internet access across the globe.

SpaceX's Starlink has been the clear leader so far, with more than 6,000 satellites already in orbit. In April 2025, Amazon launched its first batch of Project Kuiper satellites, officially entering the race. Amazon plans to eventually operate a network of more than 3,200 satellites.

These satellites orbit much closer to Earth than traditional communication satellites. Because of this shorter distance, signals travel faster and connections become more reliable for users on the ground. Traditional satellites sit roughly 35,000 kilometers above Earth, while low-orbit satellites stay within about 550 to 1,200 kilometers.

One of the biggest goals of this technology is connecting people in remote areas where ground-based internet infrastructure does not exist. Rural communities, ships at sea, and aircraft could all benefit from a reliable satellite internet connection. Some governments are also looking at satellite networks as a

backup system in case of natural disasters or network failures.

However, the growing number of satellites raises serious concerns. Astronomers warn that bright satellite trails are making it harder to observe the night sky. There are also worries about space debris — broken or retired satellites that remain in orbit and could damage active ones.

As competition in this sector grows, satellite internet may soon become a standard part of global communication. How countries manage the crowded skies above will be one of the key challenges ahead.

한글 요약

스페이스X의 스타링크와 아마존의 프로젝트 카이퍼를 비롯한 여러 기업들이 수천 개의 저궤도 위성으로 구성된 메가콘스텔레이션 구축 경쟁을 벌이고 있다. 저궤도 위성은 기존 위성보다 지구에 훨씬 가까운 550~1,200킬로미터 고도에서 운용되어 빠르고 안정적인 인터넷 연결을 제공하며, 오지 지역과 재난 상황에서의 통신 수단으로도 주목받고 있다. 그러나 위성 수의 급증은 천문 관측 방해와 우주 쓰레기 증가라는 심각한 문제를 초래하고 있다. 위성 인터넷이 글로벌 통신의 표준으로 자리잡아 가는 가운데, 각국이 혼잡해지는 지구 상공을 어떻게 관리할지가 향후 핵심 과제로 부상하고 있다.

WORDS & MEANINGS

megaconstellation 초대형 위성군
orbit 궤도를 돌다; 궤도
infrastructure 기반 시설
debris 잔해, 파편

remote 외딴, 멀리 떨어진
reliable 신뢰할 수 있는, 안정적인
altitude 고도

Robots Step In to Care for the Elderly

By Daniel · 2026-07-29 · 262 words · L2



고령화 사회와 로봇 돌봄

Populations around the world are aging rapidly. In many countries, the number of older adults is rising while the number of younger workers continues to fall. This creates a serious problem: who will care for elderly people when there are not enough caregivers?

Robots are increasingly seen as part of the answer. Care robots can help older adults with daily tasks such as lifting, moving around, and taking medication on time. Some robots also monitor a person's health and send alerts to medical staff when something seems wrong.

Social robots are another growing category. These devices are designed to reduce loneliness by holding simple conversations and responding to emotions. Experts say loneliness is a major health risk for elderly people. This means emotional support can be just as important as physical care.

However, robot care raises serious ethical questions. Critics worry that replacing human contact with machines could make older adults feel undervalued. There are also privacy concerns, since care robots often collect personal data.

Countries such as Japan and South Korea are leading the way

in developing care robots. Both nations face some of the world's fastest-aging populations. Japan, for example, has encouraged hospitals and care homes to test robot technology as a response to its growing care shortage.

The debate is not just about technology, though. It is also about what kind of care society truly values. As robot care becomes more common, governments and families will need to decide how to use it responsibly. At the same time, they must ensure that older adults still receive genuine human connection.

한글 요약

전 세계적인 고령화로 돌봄 인력이 부족해지면서, 신체 보조와 건강 모니터링을 제공하는 돌봄 로봇이 주목받고 있다. 소셜 로봇은 노인의 고독감을 줄이기 위해 대화와 감정 반응 기능을 갖추는 등 정서적 지원 역할도 담당한다. 그러나 인간적 접촉의 대체와 개인 데이터 수집에 따른 윤리적·프라이버시 문제도 제기되고 있다. 일본과 한국 등 고령화 선진국들이 로봇 기술 도입을 이끄는 가운데, 기술의 책임 있는 활용과 노인의 진정한 인간적 유대 보장이 핵심 과제로 떠오르고 있다.

WORDS & MEANINGS

caregiver 돌봄 제공자, 간병인
medication 약, 약물
monitor 감시하다, 관찰하다
ethical 윤리적인

privacy 사생활, 개인 정보 보호
trial 시험하다, 시험적으로 사용하다
aging 노령화, 나이 드는

Europe's Carbon Tax Reshapes Global Trade

By Daniel · 2026-07-29 · 264 words · L2



탄소 국경세가 무역을 바꾼다

When a factory in China produces steel and sells it to a European buyer, the price used to ignore the pollution created in making it. That is now changing. The European Union's Carbon Border Adjustment Mechanism, known as CBAM, charges importers a fee based on how much carbon dioxide was released during production. The policy took effect in its transition phase in October 2023 and will be fully enforced from 2026.

The core idea behind CBAM is fairness. EU companies already pay for their carbon emissions under the bloc's internal carbon pricing system. Without a border charge,

imported goods made with cheap, polluting energy would have an unfair price advantage. CBAM is designed to close that gap.

The sectors affected first include steel, cement, aluminum, fertilizers, hydrogen, and electricity. These are all industries where production methods vary widely between countries — and where carbon emissions can be very high. Exporters will need to report the emissions linked to their goods, and importers will pay for any difference between what was paid abroad and the EU carbon price.

Developing countries face a particular challenge. Many nations in Africa and Southeast Asia rely on fossil fuels for manufacturing and lack the resources to upgrade quickly. Researchers warn that CBAM could hurt poorer exporters more than richer ones, creating an unequal burden in global trade.

More than 40 countries are now considering similar carbon border measures, which suggests this approach could become a global standard. For students watching international economics, CBAM is a clear example of how climate policy and trade policy are becoming impossible to separate.

한글 요약

EU의 탄소국경조정제도(CBAM)는 제품 생산 과정에서 배출된 탄소량에 따라 수입업자에게 비용을 부과하는 제도로, 2023년 10월 전환기 시행을 시작하여 2026년 전면 적용될 예정이다. 이 제도는 이미 탄소 비용을 부담하는 EU 기업과 저비용 오염 에너지를 사용하는 해외 생산업체 간의 불공정한 경쟁을 해소하기 위해 도입되었다. 그러나 화석연료 의존도가 높고 시설 현대화 여력이 부족한 아프리카·동남아시아 등 개발도상국에 불균형한 부담을 줄 수 있다는 우려도 제기된다. 현재 40개국 이상이 유사한 제도를 검토 중인 가운데, CBAM은 기후 정책과 무역 정책이 긴밀하게 연결되어 가는 국제 경제의 새로운 흐름을 보여주는 대표적 사례로 주목받고 있다.

WORDS & MEANINGS

mechanism 제도, 체계

emission 배출, 배출량

importer 수입업자

enforce 시행하다, 집행하다

fossil fuel 화석 연료

carbon pricing 탄소 가격제

transition 전환, 과도기

burden 부담

The Silent Pandemic Hiding in Plain Sight

By Daniel · 2026-07-29 · 139 words · L1



항생제 내성, 조용한 팬데믹

Antibiotics have saved millions of lives, but their power is fading. When bacteria are exposed to antibiotics too often, they can change and become resistant. These resistant bacteria are sometimes called "superbugs."

Antimicrobial resistance, or AMR, happens when bacteria, viruses, or fungi stop responding to medicines. This can turn common infections — like pneumonia — into serious, hard-to-treat illnesses. Health experts warn that AMR is spreading quietly worldwide, earning it the name "silent pandemic."

One major cause is overuse of antibiotics in medicine and agriculture. When farmers give antibiotics to healthy animals, resistant bacteria can develop and spread through food. Reducing unnecessary antibiotic use is one of the most important steps to slow this threat.

Developing new antibiotics is slow and expensive. Simple actions — like finishing a full course of antibiotics — can still make a real difference.

한글 요약

항균제 내성(AMR)은 세균·바이러스·진균이 의약품에 반응하지 않게 되는 현상으로, 흔한 감염병을 치료 불가능한 중증 질환으로 만들 수 있다. 의료와 축산 분야에서의 항생제 남용이 주된 원인으로 지목되며, 내성 세균은 식품을 통해 광범위하게 전파될 수 있다. 보건 전문가들은 AMR이 '침묵의 팬데믹'으로 불릴 만큼 전 세계적으로 조용히 확산되고 있다고 경고한다. 새로운 항생제 개발이 더디게 진행되는 상황에서, 항생제를 처방대로 끝까지 복용하는 등의 개인적 실천이 내성 확산을 억제하는 데 중요한 역할을 한다.

WORDS & MEANINGS

resistant 저항하는, 내성이 있는

antibiotic 항생제

bacteria 박테리아, 세균

infection 감염

agriculture 농업

pandemic 전 세계적 유행병

overuse 남용, 과도한 사용

The Hidden Water in Your Food

By Daniel · 2026-07-29 · 121 words · L1



글로벌 물 부족과 가상수 무역

Water is becoming scarce in many parts of the world. About two billion people currently live in areas where water is difficult to find, and demand is growing every year.

When countries trade food, they are also trading what experts call 'virtual water' — the water used to grow or produce that food. For example, producing one kilogram of beef requires roughly

15,000 liters of water. When a dry country imports beef, it saves its own water supply.

However, research shows a troubling pattern. Water-scarce countries often export food to water-rich ones, which means they are losing water they can barely afford. Experts say better global trade policies could help fix this imbalance and protect the world's most stressed water supplies.

한글 요약

전 세계적으로 물 부족 현상이 심화되면서 현재 약 20억 명이 물 접근이 어려운 지역에 살고 있다. 국가 간 식량 교역에는 해당 식량 생산에 사용된 '가상수'가 함께 이동하며, 소고기 1킬로그램 생산에만 약 15,000리터의 물이 소요된다. 그러나 실제로는 물 부족 국가들이 물이 풍부한 국가들로 식량을 수출함으로써 오히려 자국의 수자원을 소모하는 역설적인 현상이 나타나고 있다. 전문가들은 이러한 불균형을 해소하기 위해 글로벌 무역 정책의 개선이 필요하다고 강조한다.

Source · ncbi.nlm.nih.gov

WORDS & MEANINGS

scarce 부족한, 희귀한

demand 수요, 요구

virtual 가상의

import 수입하다

export 수출하다

imbalance 불균형

policy 정책

스크랩하기 Scrap It!

이름 _____ 날짜 _____ 학교 · 반 _____

1. 이번 호 기사 목록 — 스크랩할 기사에 □ 표시하세요

- ☐ 환경 Cities Around the World Are Running Out of Groundwater
- ☐ 경제 Global Race to Find Chip Engineers Heats Up
- ☐ 환경 Nuclear Power Makes a Quiet Comeback
- ☐ 세계 Nations Restrict Grain Exports, Raising Global Food Concerns
- ☐ 기술 The High-Stakes Debate Over Regulating AI

- ☐ 기술 The Race to Fill the Sky With Satellites
- ☐ 사회 Robots Step In to Care for the Elderly
- ☐ 경제 Europe's Carbon Tax Reshapes Global Trade
- ☐ 건강 The Silent Pandemic Hiding in Plain Sight
- ☐ 환경 The Hidden Water in Your Food

2. 스크랩할 기사 · 제목과 섹션

3. 기사 요약 (3문장)

4. 새로 배운 단어 5개 · 뜻

5. 내 생각 · 이 기사가 중요한 이유